SMART INDIA HACKATHON 2026

Problem Statement ID: SIH 2026 - Smart Automation (to be filled)

Problem Statement Title: Crowdsourced Road Defect Intelligence for India

Theme: Smart Automation

PS Category: Software

Team ID: To be filled

Team Name: Team SETU

SETU - Sensor-Enabled Tracking of Urban-road-damage

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The problem we are solving

9,438	**+53%**	**Rs 17,693**	**1-2**
pothole deaths in India, 2020-24	**more than five years ago**	**cost of auditing one pothole**	**checks a road gets in a year**

A survey van comes once a year and a complaint waits weeks, so no city has a live map of its own potholes.

Our solution

Sensors do 99.9% of the work. The camera does 0.1%.

1. **Phones already on the road feel the bump** — **Rs 0** a spot
   A small model on the phone decides if the jolt was a real defect. The driver does nothing.
2. **Three different vehicles must agree** — **Rs 0** a spot
   One phone is never enough. A spot counts only once three separate vehicles report it.
3. **Only the doubtful spots get a camera** — **Rs 3** a spot
   The next rider records 5-8 seconds and our vision model settles what is actually there.

What makes it different

Nothing new to install
It rides inside apps drivers already keep open all day.

Cheap first, camera last
Video is spent only where the sensors cannot decide.

The rider gets a vote
One tap, asked only while the vehicle is standing still.

A repair proves itself
When the jolts stop, the ticket closes on its own.

Private by design
Hashed device IDs, no personal data, clips gone in 7 days.

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TECHNICAL APPROACH

How the data moves, start to finish

1. Phone feels a jolt →
2. It reaches our server →
3. Nearby readings grouped →
4. Gig worker is asked to vote →
5. A short clip is asked for →
6. Confirmed or dropped →
7. Dashboard and work order

Steps 4 and 5 are the only ones that cost anything, and they run on about one spot in a thousand.

What we are building it with

On the phone	An Android app with a small AI model (TensorFlow Lite) reading the motion sensors
Our server	Python (FastAPI) with background workers doing the grouping every few minutes
Map and database	PostgreSQL with maps built in (PostGIS), so every reading sits on a real road
The dashboard	A React web map the ward officer opens in a browser, updating on its own
The AI models	YOLO looks at the clips; a small 1D-CNN reads the sensor data on the phone
Running it	Docker, GitHub and a free cloud tier - nothing here has to be bought

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FEASIBILITY AND VIABILITY

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Why the math works

We start from a published result, not from our own guess.

Wu, Liu & Ren - Sensors 20:5564 (2020)

One phone, one pass over a pothole: 88.5% of what it flags is real, and it catches 75% of the potholes.

So one phone on its own is wrong about 1 flag in 9.

We wait for 3 different phones to flag the same 20 m of road:

0.115 x 0.115 x 0.115 = about 1 false spot in 660

And three passes catch **98%** of the real potholes, not 75%.

Roughly 1 in 660 false alarms, 98% of real potholes found, and money spent only on the handful the sensors could not settle.

For SIH we will show one Android app, one server and one dashboard, tested on about 50 km of real city road.

Is it actually doable?

- Every piece is free and open source
- Runs on an ordinary Rs 8,000 Android phone
- One bus route or municipal fleet is enough to start
- Six of us: one app, one server, one dashboard

What could go wrong

Phones kill background apps
Run it as a service the user can see, and test on MIUI and ColorOS

GPS drifts by 20-30 m
Snap every reading to the road, then group inside a 20 m radius

Speed breakers look like potholes
Trained on them as 'not a pothole', and the camera overrules

Video can catch faces
Blurred on the phone and deleted after seven days

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IMPACT AND BENEFITS

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What changes on the ground

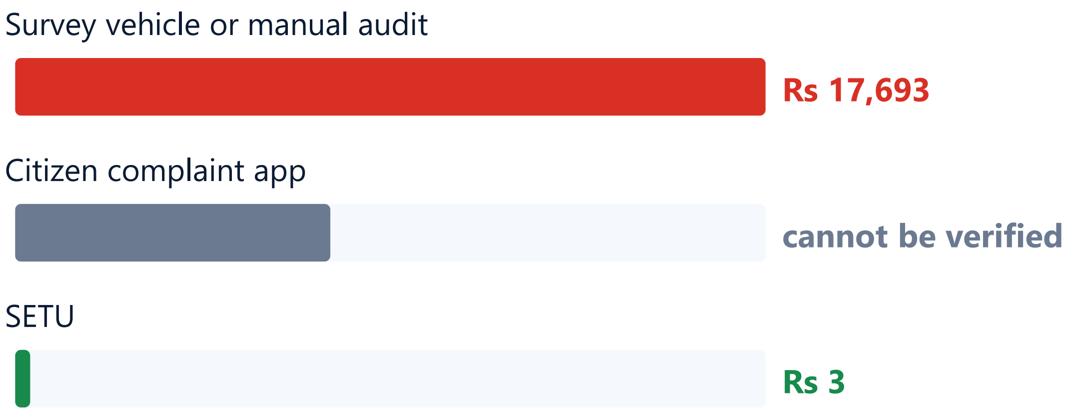
6+
DEATHS A DAY
pothole crashes, MoRTH

Daily
ROAD CHECK
instead of once a year

100%
REPAIRS CHECKED
evidence before and after

Who	Today	With SETU
Citizens	no warning, nobody answerable	**safer roads, repairs on a clock**
Municipality	costly audits, guesswork	**a ranked repair list, daily**
NHAI / PWD	one survey pass a year	**every lane that gets driven**
Gig workers	vehicle damage, back pain	**warnings ahead, no extra work**

What it costs to confirm one pothole



5,900x **cheaper for the same kind of proof, because the camera only runs when the sensors are unsure.**

Side benefits we did not plan for

- Monsoon damage shows up in days, not at the next survey
- Smoother roads mean less fuel and fewer suspension repairs
- The data shows which contractor actually delivered

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RESEARCH AND REFERENCES

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Peer-reviewed foundation

Sattar, Li & Chapman - Sensors 18:3845 (2018)

A phone's motion sensor can spot road damage; sampling rate is the real limit, so we sample at 100 Hz.

mdpi.com/1424-8220/18/11/3845

Wu, Liu & Ren - Sensors 20:5564 (2020)

88.5% precision and 75% recall from one phone - the numbers our three-vehicle math is built on.

mdpi.com/1424-8220/20/19/5564

Sensors 20:409 (2020) - participatory sensing

Measured GPS error of 27-32 m at speed, which is why we group inside 20 m after map-matching.

mdpi.com/1424-8220/20/2/409

Arya, Maeda, Ghosh et al. - RDD2022 (arXiv)

47,420 labelled road images from 6 countries including India - our training base for the vision model.

arxiv.org/abs/2209.08538

Government data and Indian standards

MoRTH - Road Accidents in India

9,438 pothole-related deaths in 2020-24, up 53% in five years.

morth.nic.in/road-accident-in-india

NHAI Network Survey Vehicles

The survey fleet covers ~20,933 km of highway about once a year, at high cost. We cover every lane driven.

nhai.gov.in

IRC:82 - Maintenance of Bituminous Surfaces

The Indian code we map severity to, so our output is directly actionable for a ward engineer.

law.resource.org - IRC 82:2015 PDF

DPDP Act 2023 + DPDP Rules (13 Nov 2025)

No personal data, hashed device tokens, consent first, 7-day media limit.

meity.gov.in - data protection framework